Chapter 3 exercises

DM03-E01

Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by the element method, and identify the use of extensionality.

DM03-E02

In $U = \{1, \ldots, 30\}$, let A, B, C be multiples of two, three, and five. Find the union size and the number belonging to exactly one set.

DM03-E03

Roll two independent fair six-sided dice. Let A mean sum seven and B mean first die four. Find $P(A), P(A|B)$ and determine independence.

DM03-E04

Under the assumed testing model, prevalence is 2%, sensitivity 95%, and specificity 90%. Use Bayes' formula for disease probability given a positive test. Give expected counts for 10,000 people and explain how the prior enters the denominator.

DM03-E05

A variable X has values 0,1,2 with probabilities $c, 2c, 3c$. Find $c, EX, \mathrm{Var}(X)$.

DM03-E06

Can positive-probability events satisfy $P(A|B) > P(A)$ but $P(B|A) < P(B)$? Prove the relation between the directions of these comparisons with marginals.

DM03-E07

Let $A = \{a, b\}, B = \{b, c\}$, with distinct symbols. List $\mathcal{P}(A) \cap \mathcal{P}(B)$ and $\mathcal{P}(A \cup B) \setminus (\mathcal{P}(A) \cup \mathcal{P}(B))$. Prove generally that $\mathcal{P}(A) \subseteq \mathcal{P}(B) \iff A \subseteq B$.

DM03-E08

For fair mutually independent bits x_1, x_2, x_3, let $A = \{x_1 = x_2\}, B = \{x_2 = x_3\}, C = \{x_1 = x_3\}$. Prove pairwise independence and decide mutual independence.

DM03-E09

Twelve distinguishable components contain four defects. Uniformly sample three without replacement. Find the complete PMF, mean, and variance of defect count X. Recompute variance using covariances of three position indicators.

DM03-E10

On uniform $\Omega = \{00, 01, 10, 11\}$, let I_1, I_2 be the bits and I_3 their XOR. Find the PMF, mean, and variance of $S = I_1 + I_2 + I_3$, comparing with Binomial$(3, 1/2)$. Do equal means and variances prove equal distributions?

DM03-E11

Suppose $P(Y = 0) = 2/3, P(Y = 1) = 1/3$, with $X|Y = 0$ Bernoulli$(1/4)$ and $X|Y = 1$ Bernoulli$(3/4)$. Find $P(X = 1)$, then both within-group and between-group terms of total variance, checking directly.

DM03-E12

Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Find covariance, determine independence, and compute $\mathrm{Var}(X + Y)$. Explain the logical difference between zero covariance and independence.

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DM03-E13

Let first-success trial T be Geometric$(1/3)$, supported from one. Find mean, variance, and $P(T \geq 9)$, comparing Markov and Chebyshev bounds for this one-sided tail. Prove memorylessness $P(T > m + n|T > m) = P(T > n), m, n \in \mathbb{Z}_{\geq 0}$.

DM03-E14

Joint masses at $(0, 0), (0, 1), (1, 0), (1, 1)$ are $1/8, 3/8, 3/8, 1/8$. Find marginals, $P(X = 1|Y = 1)$, covariance, and $\mathrm{Var}(X + Y)$. Check the sum's PMF.

DM03-E15

On $\Omega = \{1, 2, \ldots\}$, let $P(\{n\}) = 1/[n(n+1)]$ and $X(n) = n$. Prove this is a probability distribution, find $P(X \geq m)$ for positive integers m, and prove EX is not finite. Why is $E[X^2] - (EX)^2$ unavailable as a finite-variance calculation?

DM03-E16

On $\Omega = \{a, b, c\}$, masses are $1/2, 1/2, 0$. Put $A = \{a\}, B = \{c\}$. Is B empty? Is $P(A|B)$ defined here? Are A, B independent? Apply definitions, distinguishing sets and probabilities rather than relying on everyday "impossibility."

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 3.

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